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ORIGINAL PAPER

Optimizing harvesting for facial lipografting with a new photochemical stimulation concept: One STEP technique™

Patricio Centurión[1] · Ronald Gamarra[1] · Gonzalo Caballero[1] · Paul Kaufmann[1] · Pia Delgado[1]

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Abstract

Background Facial fat grafting for rejuvenation is one of the most popular facial aesthetic procedures in plastic surgery. It is always challenging and since there are a lot of techniques for adipose tissue (AT) harvesting, there are no standard procedures that guarantee natural and long-lasting results. We developed the selective tissue engineering photo stimulation technique (One STEP™) in which we used a novel infrared 1210-nm wavelength laser diode for fat preserved harvesting and direct fat injection that we named PicoGraft™, with no fat manipulation.

Methods This is a retrospective descriptive study in which we included all senior author's patients that got facial fat grafting using the One STEP™ technique. We compared the AT aspirated, after laser emission (STEP-PicoGraft) and the standard assisted liposuction samples (SAL) in cultures. We study the mitochondrial activity of the ASC between STEP and SAL in fresh samples and after 24 h. The evaluation of the results included subjective changes regarding wrinkles, grooves, palpebral bags, hyperchromic spots, and fat hypotrophy of our patients.

Results Between July 2013 and May 2018, a total of 245 patients underwent facial fat grafting using this novel technique. We observed adipocytes preserved after STEP harvesting comparing morphologic changes in SAL samples with a high concentration of inflammatory particles in cultures. ASC mitochondrial activity shows an important difference of more than 7 times in STEP samples in fresh analysis that increase 12 times in 24 h. The subjective results show a good improvement in the periorbital area. The changes on the skin and subcutaneous tissue are seen from the second month and continue to improve up to 12 months.

Conclusions Facial fat grafting using the PicoGraft™ obtained by One STEP™ technique gives excellent volumetric and regenerative results in a single treatment without volumetric hypercorrection, and it is a good alternative for facial rejuvenation. The fat graft obtained with this novel technique is homogenous, without lumps, and has high concentration of viable stimulated ADSC and a high number of viable adipocytes.

Level of evidence: Level III, therapeutic study.

Keywords Fat graft · Lipolaser · One STEP · Facial rejuvenation · PicoGraft

Introduction

Surgical facial rejuvenation has always been challenging for plastic surgeons. The final goal is to obtain natural and long-lasting results. The aging process affects not only the skin but also all the other facial structures; there is a hypotrophy of the underlying fat tissue, flaccidity of facial muscles, and facial bone reabsorption [1].

In the past decades, there have been several techniques to restore and rearrange the fat tissue that has been lost over time. One of these techniques is the grafting of autologous fat. There is a lot of controversy on how to harvest and treat the fat in order to optimize long and lasting results. Many surgeons recommend aspiration, decanting, and immediate fat grafting, taking into account that the least traumatic the procedure, the better and long-lasting result. Therefore, we believed that the quality of the fat would be operator dependent [2–4].

Recent studies about the fat harvested for facial lipofilling have shown that besides adipocytes, we also aspirate adipose-derived stem cells (ADSC), capable of differentiating into any cellular line. These findings suggest that fat tissue could provide the main elements for soft tissue regeneration [5–7], and several studies have already been done in many medical specialties [8, 9]. This is why lipofilling is considered a single treatment capable of providing and restoring both of the lost

✉ Patricio Centurión
pccenturion@gmail.com

1 Universidad Peruana de Ciencias Aplicadas, Lima, Peru

volume and regeneration of soft tissues, giving the patient a better overall aesthetic results.

Today the "gold standard technique" for fat harvesting is the standard assisted liposuction (SAL). The high incidence of fat reabsorption after lipografting suggested that it could be because of cell damage [10, 11]. Our concern was then how much cell damage is performed by the standard assisted liposuction (SAL). Our concept is that SAL with its "mechanical disruption" method works like a lipo-curettage, responsible for the damage of all cellular components in the subcutaneous cellular tissue.

In order to obtain a better adipocyte quality as well as all other cellular lines for facial volume restoration that could have better survival and long-lasting results, we started to look for other harvesting alternatives.

The laser-assisted liposuction (LAL) technique describes adipocyte lysis by increasing the cytoplasmic membrane permeability. This is caused by the photothermic laser property. The lipolysis allows for an easier fat extraction but limits its use as a fat graft.

The laser effect on cells depends on multiple factors such as the wavelength used, the specific target each one has, and the time of laser application as well as the energy employed (fluency) [12].

In 2010, we started to evaluate some of these variables to see if we could obtain better fat grafting material.

We developed the selective tissue engineering photo stimulation technique (One STEP™) in which we used a novel infrared 1210-nm wavelength laser diode with fat-rich tissue as target with the specific parameters. This technique uses more of the photochemical/biomodulation properties of laser rather than the photothermic/lysis properties found in most current laser platforms.

We performed many laboratory studies in which we compared several variables of the adipocytes and ADSC obtained by SAL and those samples obtained by the One STEP technique. The variables measured and compared were adipocyte histology, the number and percentage of ADSC's viability in both fresh and cultured samples, mitochondrial activity (MTT, in both fresh samples and after 12 h of culture), and finally the temperature of the skin and subcutaneous tissue before/after infiltration and immediately after the laser application. In Figs. 1 and 2 the different histologies are compared and described. In Table 1, we show our comparison in mitochondrial activity.

We performed comparative analysis in the SAL and One STEP samples measuring the ASC mitochondrial activity in time, showing 7.11 times more in One STEP in fresh samples and an increase difference in favor of One STEP of 12.98 times always versus the SAL samples. In SAL samples, the apoptosis can explain this behavior and in the One STEP samples, the photo stimulation property of the laser 1210 nm.

Our objective with this paper is to present an innovative technique in fat harvesting that provides better grafting material, the PicoGraft™, for facial far graft. The innovative part consists of the subcutaneous application of infrared laser energy with a novel 1210-nm laser wavelength. This infrared energy produces denaturalization of the connective tissue releasing adipocytes and ADSC, while preserving adipocytes and selectively stimulating the ADSC; hence, it was named selective tissue engineering photostimulation (One STEP.) This procedure requires just one step, without any manipulation or processing of the subcutaneous aspirated tissue, and it can be immediately grafted [13–15].

Materials and methods

This is a retrospective descriptive study in which we included all patients that got facial fat grafting using the One STEP™ technique created by the senior author.

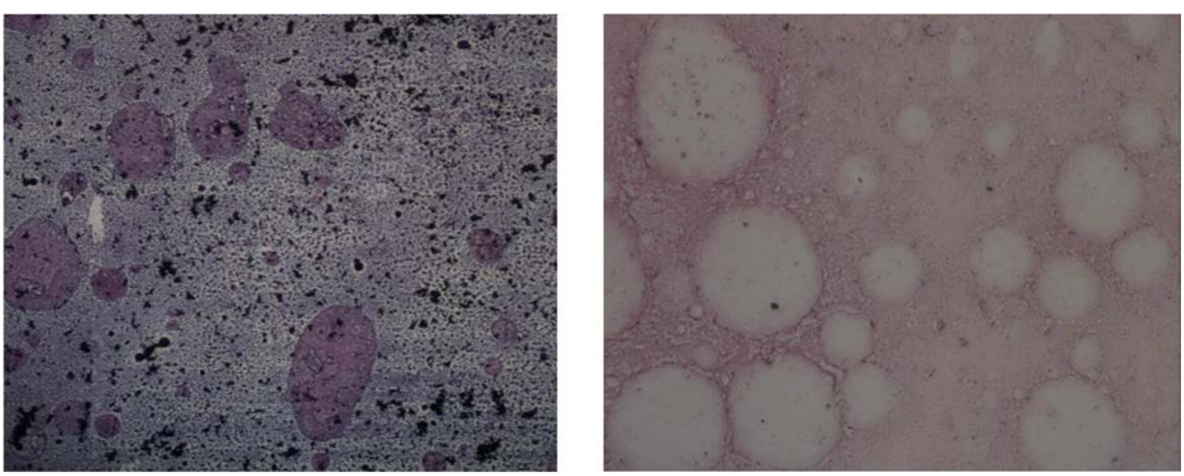


Fig. 1 **Microscopy 20x: fat cells, 12 hours after (AT) harvest**: We can see the differences between SAL (A image) and One STEP™ (B image), whereas the histology is preserved with no inflammatory particles (black spots) compared with the SAL where the adipocytes present an altered morphology, detritus (cytoplasmic membrane) and a high number of inflammatory particles

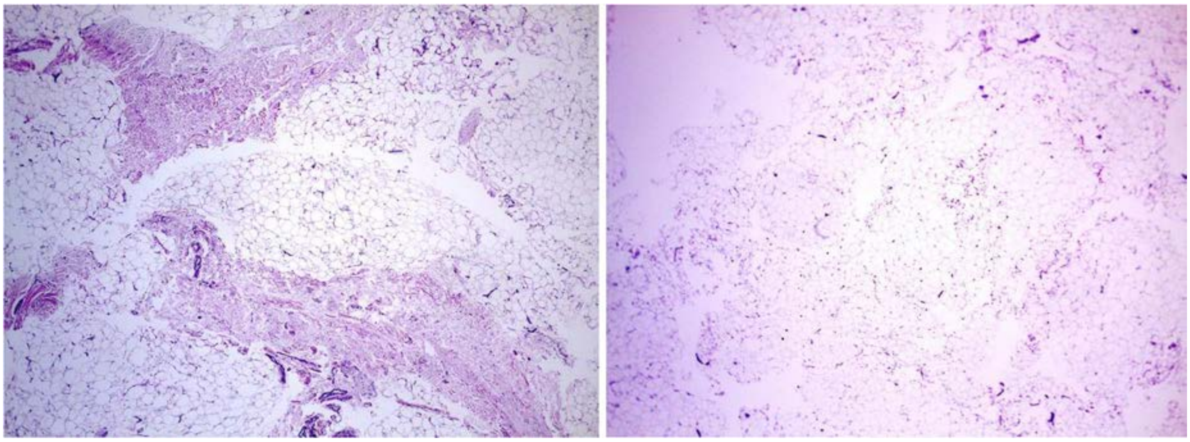


Fig. 2 **Connective Tissue** We can see in these comparative images of the AT (syringe) aspirated fresh samples the differences of the amount of connective tissue between SAL (LEFT) which is the dense pink color surrounding fat cell whereas in the One STEP (RIGHT) is almost nonexistent (hematoxylin/eosin)

An infrared 1210-nm laser diode was used (DMC–Brazil) with the “ASC harvest” presetting. The evaluation of the results included subjective changes regarding wrinkles, grooves, palpebral bags, hyperchromic spots, and fat hypotrophy. Pretreatment and post-treatment photos were taken at 3, 6, 9, and 12 months.

One STEP™ harvest technique

The procedure is done under IV sedation. Peri-umbilical markings are done where the adipocytes and ADSC will be obtained. We use a solution of cold saline (4 ° C^0 + adrenaline 1:500,000 + lidocaine 0.35%) and infiltrate it at a deep level over the fascia superficialis using a 2 mm Klein cannula (wet technique). Especial consideration taken is not to tunnelize the subcutaneous cellular tissue so that the histologic architecture is preserved (Yellow Arrow), as we can see in Fig. 3.

The next step is the application of the 1210-nm laser energy that has greater affinity/absorbance for adipose tissue. The laser is applied at all levels of the subcutaneous cellular tissue using a 600-µm fiber optic inside a 2-mm cannula. The tip of the fiber optic should protrude 3 mm at the tip of the cannula. The application is performed by gentle back and forward movements achieving a homogeneous histologic architecture as we can see in Fig. 4 where all fibrous septae are gone (Green Arrow).

Table 1 Comparative Mitochondrial Activity ad fresh and 24 hours samples

Samples	Point Zero (10 min)	24 hours	
A1	0.686	0.652	Conventional
A2	0,686	0,670	Conventional
B1	4.880	8.445	Laser 1210-nm
B2	4.880	8.895	Laser 1210-nm
MTT	0.623	0.600	CONTROL
MTT	0.623	0.623	CONTROL

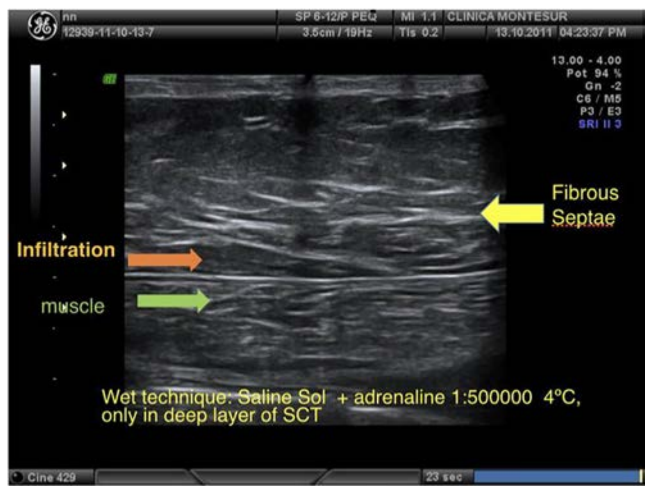


Fig. 3 Ultrasound image capturing deep plane infiltration

Fig. 4 Ultrasound image capturing deep plane infiltration

Fig. 5 Diagram of the access points for the zones to be lipoinjected. At the infraorbital, supraorbital, chin, and marionette levels, micro cannulas (1.2 mm) are used, curves and straight lines. In the glabellar, temporal, and nasogenian regions, straight micro cannulas were used. For the cervical area, 23G needles were required

Then, we perform the aspiration of the subcutaneous tissue using 3.5-mm cannulas with 50 cc syringes, and the fat tissue harvested is transferred into 10 cc syringes where they remain ready for grafting. This final product we named PicoGraft™.

It is a closed circuit, where no manipulation is performed and nothing is discarded.

Facial lipografting technique

The harvested fat tissue is grafted using a straight and curved 1.2-mm microcannulas and a #23-gauge needle in a 1 cc syringe. The point of entry is done with a #18-gauge needle after injecting a small amount of local anesthetic. The area to be treated is not infiltrated. Access points are shown in Fig. 5.

The graft is placed at a deep level, respecting the subdermic fascias of each area. It is placed with slow retrograde movements until the desired volume is achieved. We can appreciate this deep plane in the next ultrasound image (Fig. 6).

No overcorrection is made and tunnels do not cross each other. Finally, paper tape is placed on the entry points. When a facelift is performed, we prefer the short scar facelift technique [16], and the fat grafting is performed at the end of the procedure. Only prophylactic antibiotics are used.

Results

Between July 2013 and May 2018, 245 patients underwent face fat grafting using the One STEP™ technique. Demographic data as well as the diagnosis is shown in Table 2. The areas and media volumes grafted are shown in Fig. 7. Associated procedures are listed in Table 3.

The subjective results show a good improvement in the periorbital area (supra and infraorbital and glabellar area).

Fig. 6 Deep Fat Grafting Injection, where the white arrow shows the superficial dermis and the green arrow the supraperiosteal plane next to the straight cannula (red arrow) depositing fat graft at the end of the cannula (blue arrow)

Table 2 Demographic data and diagnosis of the 245 patients who underwent face fat grafting using the One STEP™ technique

Patients		245
Sex		
Female		226 (92.24%)
Male		19 (7.76%)
Average age		49.13 years
Preoperative diagnosis		
Facial	225	91.8%
Facial lipodystrophy	3	1.22%
Facial hipotrophy	3	1.22%
Hypomentonism	2	0.81%
Mixture	4	1.63%
Others	5	2.1%

Table 3 Esthetic procedures associated

Liposculpture	98	40%
Lateral blow lift	66	26.93%
Cervical lipolaser	41	16.735
Mandibular lipolaser	26	10.61%
Blepharoplasty	38	15.51%
Full facelift	26	10.61%
1/3 inferior facelift	16	6.53%
Rhinoplasty	7	2.85%
Breast augmentation	11	4.48%
Breast lipografting	9	5.32%
Lipoabdominoplasty	10	4.08%
None	15	6.12%
Others	2	0.81%

Photos were taken at 0, 4, 6, and 9 months post-surgery. Representative cases are presented in Figs 8, 9, 10, and 11.

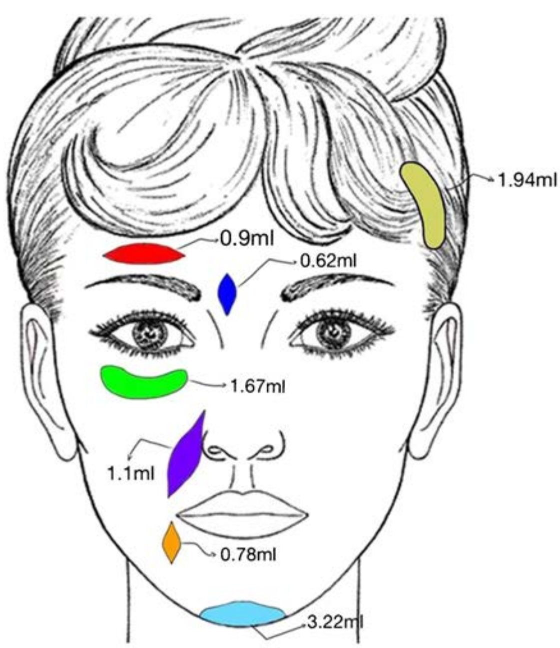
Case 5 shows a 52-year-old patient who received a total of 13 cc autologous One STEP facial fat grafting. We can appreciate in the before pictures frontal and glabellar wrinkles, skin sun damage, hypotrophy of the infraorbitary area with nasojugal grooves. In nine-month post-surgery, we can see a good improvement on wrinkles without the need of botulinic toxin and natural volumetric preservation of fat. We add the markings generally used for facial fat grafting for all our patients (Fig. 12).

Only 6 complications were seen (2.4%) which were late edema and persistent edema. All cases resolved over time, aided with the application of infrared laser to facilitate lymphatic drainage.

Discussion

Currently, facial lipografting is a very common technique used for volume repositioning. Several techniques are being studied to optimize facial lipofilling, seeking natural results with long-standing results and the regenerative effect of the ASC over facial tissues [17].

SAL with its "mechanical disruption" mechanism continues to be the most popular method for fat and subcutaneous tissue harvesting. There are several protocols available, to treat/manipulate the aspirated tissue. As shown by our previous histologic comparative studies, the SAL samples after 12-h culture



AREA LIPOINJECTED	AVERAGE VOLUME IN ML
INFRAORBITAL	1.67
NASOGENIAN GROOVE	1.10
MARIONETTE LINE	0.78
GLABELLAR	0.62
SUPRAORBITAL	0.90
MANDIBULAR SIDE	1.83
CHIN	3.22
TEMPORAL	1.94
SUPERIOR ORBITAL RIM	0.33
UPPER LIP	0.83
INFERIOR LIP	0.66
SUPRALABIAL GROOVES	0.50
*OTHER ZONES (CAPILAR, cervical, ETC)	Máx. 30.6
	Min. 0.2

Fig. 7 Diagram of the access points for the zones to be lipoinjected. At the infraorbital, supraorbital, chin, and marionette levels, micro cannulas (1.2 mm) are used, curves and straight lines. In the glabellar, temporal, and nasogenian regions, straight micro cannulas were used. For the cervical area, 23G needles were required

Fig. 8 **Case 1** shows a woman of 32 years old, who presented a mild nasolabial fold and nasojugal fold due to fat hypotrophy (A). She had an under-projected chin and showed a dull skin (C). She was lipoinjected with 15 ml of fat graft using One STEP technique. 4-month postoperative results show an improvement in skin quality with a brighter appearance (B). There's a decrease in both nasolabial and nasojugal folds, and more anteriorly projected chin (B, D)

showed a great number of destroyed or morphologic altered adipocytes, adipocyte cytoplasmic membrane fragments, and inflammatory particles, due to mechanical cannula disruption. In 2010, we used a 1210-nm diode laser prototype for the first time, and we showed that by a connective tissue denaturalization mechanism, it produces a change of the histochemical properties, liberating adipocytes, with preservation, 98% of them and with no inflammatory molecules, unlike the SAL samples [13, 18, 19].

In addition, centrifuging, decanting, and washing the samples of harvested tissue only increase manipulation, air exposure, and cellular trauma which increases cell lysis and apoptosis, although Conde concludes that only sample washing increases cell viability for grafting [20]. Another simple difference is the amount of infranadant between SAL and LAL, as its shown in Fig. 13.

Infranadant comparison: SAL (left) and One STEP (right)

Face lipografting can be associated with other procedures such as facelift, blepharoplasty, and upper third face surgeries, providing better results rather than tissue repositioning alone [21].

The One STEP technique also provides the benefit that the ADSC, found in the harvested and grafted samples, has on tissue regeneration.

Fig. 9 **Case 2** shows a 44 years old woman with a moderate tear trough deformity associated with periorbicular hyperpigmentation (A). She showed a flat malar region with a negative vector, dull skin, with perinasal and peribuccal telangiectasia (C). She was lipoinjected with 10 ml of fat with the One STEP technique. 9-month postoperative results show a correction of tear trough deformity and a more projected malar region (B, D). There's an improvement in skin quality, showing a smoother and brighter skin, and an attenuation of periorbicular hyperpigmentation (B, D)

Fig. 10 **Case 3** shows a female patient 29 years old with an hipotrophy in the malar region, nasojugal grooves and periorbicular oris hyperpigmentation (A, C). Two months after of facial One Step fat grafting we can see the volumes are preserved and quality skin changed with brightness and improvement of acne sequels as well (B, D)

Our proposal is based on a study we initiated in 2006, searching for a technology for liposculpting that would preserve the subcutaneous tissue and not destroy it. The light/laser technology has many different properties such as the photothermic, photo acoustic, photochemical, and photomechanical that are not very well-known.

We based our technique in the photochemical property of laser, the same that happens in photosynthesis, where luminic energy (photons) is transformed into stable chemical energy without having the photothermic effect [18].

Other technologies such as ultrasound (Vaser TM) have been used trying to obtain adipocyte and ADSC preservation for grafting. However, comparative studies between ultrasound and SAL show similar characteristics of the tissue harvested [22]. In addition, ultrasound has some complications such as burns and seromas caused by adipocyte destruction due to the temperatures obtained.

We agree with those authors in that the results of ultrasound and SAL samples are similar because both are based on a mechanical/physical action that finally destroy the adipocytes; something that does not happen with our technique as shown on our previous studies [23].

In the One STEP technique, a main photochemical reaction is produced, generating a direct activation on collagenase endogenous enzymes so as in the cytochrome c oxidase, being the ADSC mitochondria of the first receptor.

Fig. 11 **Case 4** shows a women 34 years old with periorbitary hypotrophy most predominant in the nasojugal groove with hiperchromic skin changes (A, C). 2 months after her facial fat grafting in the infraorbitary area we can see a better skin quality (reduced skin hyperpigmentation) and a well balanced nasojugal groove (B, D) with harmony of the periorbital fat of the lower lid given the impression that has been taken out, which it have never

Fig. 12 **Case 5** shows a 52 year old patient who received a total of 13cc autologous One STEP facial fat grafting. We can appreciate in the before pictures (A, D) frontal and glabelar wrinkles, skin sun damage, hypotrophy of the infraorbitary area with nasojugal grooves. 9 months post surgery we can see an good improvement on wrinkles without the need of toxin botulic and natural volumetric preservation of fat (C, F) . We add the markings generally used for facial fat grafting for all our patients (B, E)

We based our investigation on Anderson's and Wassmer's work that described certain laser wavelengths that had more affinity for fat-rich tissues [24, 25].

The One STEP technique is based primarily on the photochemical property and secondary on the photothermic property, which does not traduce into a big temperature change in the skin or the subcutaneous tissue, measuring 31 °C and 30 °C respectively. Changes in skin and subcutaneous temperatures were measured and are shown in Fig. 14.

Therefore, there is no thermolysis effect like the one produced by the 980, 1064, and 1440-nm laser wavelengths as well as the ultrasound (VASER).

In 2011, we were able to confirm a higher number of viable and healthy ADSC cells; using a mitochondrium activity test (MTT) in fresh samples, we obtained an index of 0.686 in SAL versus 4.880 in the One STEP. After a 24-h culture, the index changed to 0.652 for SAL samples and to 8.445 for the One STEP (See Table 2).

This was due to an apoptosis of the ADSC with SAL. The One STEP technique, by its connective tissue denaturalization mechanism, liberates the ADSC without damaging and even more the laser emission stimulates ADSC's mitochondrium.

Fig. 13 Infranadant comparison: SAL (left) and One STEP (right)

TEMPERATURE

INFILTRATION: SALINE 4 Celsius wet technique

SAFE RANGE OF TEMPERATURES IN TISSUES		
1210 nm Laser - Wavelenght		
Surgical Time	Temperature (°C)	
	Skin	Subcutaneous Tissue
Pre-Infiltration	31.5°	33°
Post-Infiltration	30°	27°
Post-Laser	31°	30°
Post-Aspiration	30°	29.5°

Photothermic effect : Minimally

Fig. 14 Changes in skin and subcutaneous temperatures

During the SAL technique, the tissue is subjected to more manipulation (centrifugation, decantation, and filtration) that increases cellular injury that could affect the survival of adipocytes and ADSC.

The material discarded after decanting, centrifuging, or filtering could be another factor contributing to the heterogeneous results obtained with SAL, but Francis presented an enhanced method for isolating and quickly expanding a robust population of mesenchymal stem cells (MSCs) derived from lipoaspirate. This isolation process yields an abundant population of ADSCs (~ 100,000 cells per 100 ml of blood/saline collected from lipoaspirate) [26]. This is usually discarded in the SAL technique.

Our proposed technique consists of a single one-step procedure, so we consider it as a minimal-grade manipulation (MGM) [27] without processing, keeping it in a closed system and preserving all the regenerative elements of subcutaneous tissue. By using the adequate infrared laser, we are able to denaturalize the connective tissue and preserve adipocytes and ADSC, as well as stimulate them. Because we use a minimal-wet infiltration, the harvested material we obtain has no infranatant and it is homogenous and has no lumps. This allows us to inject with micro cannulas without the need to homogenize the sample harvested.

Overall, the One STEP technique provides us with a higher quantity and quality of ADSC when compared with SAL samples [18]. However, we can mention some drawbacks we found performing this technique and that many surgeons could face applying STEP, such as:

– The device is not commercialized over the world today.
– Additional time for the laser can make surgery last a little longer.
– This is a different protocol so technically, it can be a little difficult for the surgeon who is used to SAL to change for a minimal infiltration and laser application in deep planes.
– Despite above, STEP needs a (short) learning curve to obtain the same results.

The regenerative and volume repositioning effect of the ADSC depends on their number and stimulation as well as the presence of viable adipocytes. Therefore, we can obtain better homogenous aesthetic results in short periods of time for long-lasting results. The new practice of the plastic surgery is to offer volumetric and regenerative treatment, so if we compare the PicoGraft with the Nanograft, we observe that the PicoGraft offers viable adipocytes and ADSC stimulated by laser without manipulation, compared with Nanograft that only offers ADSC, in less number. Tonnard obtained an ADSC count of 1.9–3 × 10^6/100 ml of lipoaspirate [28]. With One STEP technique, we obtain 49 × 10^6/100 ml of lipoaspirate [19].

Conclusions

- Facial lipofilling using the PicoGraft™ obtained by One STEP ™ technique gives excellent volumetric and regenerative results in a single treatment without hypercorrection and is a good alternative for facial rejuvenation.
- The graft obtained is homogenous, without lumps, and has high concentration of viable stimulated ADSC and a high number of viable adipocytes, as demonstrated in our results.
- The PicoGraft™ by the One STEP™ technique has minimal manipulation, preserves all subcutaneous components, and is performed in a closed circuit which favors cell quality, quantity, and survival.
- Not all surgical laser devices destroy fat tissues, as it is commonly believed. On the other hand, we can take advantage of the not well-known photochemical property of 1210-nm laser that can selectively stimulate the ADSC and preserve adipose cells, opening this new door for regenerative medicine.

Compliance with ethical standards

Conflict of interest Dr. Patricio Centurion has ownerships of both One STEP technique and PicoGraft technique.

Dr. Centurion has technical support of DMC Laser-Brazil. Dr. Centurion is the owner of the One STEP™ technique and PicoGraft™. The rest of the authors do not have any conflict of interest to declare.

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Ethical approval This study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards. If doubt exists whether the research was conducted in accordance with the 1964 Helsinki Declaration or comparable standards. However for this retrospective study formal consent is not required.

Informed consent Patients provided written consent before their inclusion in this study. Additional consent was obtained for the use of their images.

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